$\mu \to e\gamma$ in the S_3 3HDM: forbidden at the exact vacuum, and the first LFV constraint that bites on the scalar sector

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Abstract

I have computed $\mathcal{B}(\mu \to e\gamma)$ at all 2028 viable points of the soft-broken scan, for every allowed value of the lepton parameter μ_3^ℓ and every branch of the charged-lepton fit. That is 417 992 configurations, compared with the MEG II limit 1.5×10^{-13} [1]. The amplitude sums the one-loop diagrams of all seven physical scalars and the Barr–Zee two-loop diagrams with top, bottom, τ, W and charged-Higgs loops. Each piece is checked against an independent source. Three results stand out. First, at the exact-S_3 vacuum the rate is *identically zero*: a residual $\mathbb{Z}_2$ forbids it to all orders, so $\mu \to e\gamma$ measures the soft breaking directly. Second, in the soft vacuum it is by far the strongest LFV constraint. Only **618 of 2028** scalar points keep an allowed lepton configuration, where the Higgs LFV decays keep 2023. Third, the W-loop Barr–Zee diagram through the SM-like Higgs dominates, so the survivors are the points closest to the alignment limit. The conclusion survives the two weakest parts of the amplitude, the quark couplings and a 14% normalization difference in the W loop.

1 Summary

Everything below comes from `research/thdm_s3/07_mu_to_e_gamma.ipynb`, with the numerics in `radiative.py`, `soft_decays.py` and `lfv.py`.

1. **The couplings** of every scalar come off the Lagrangian and agree with `feynlag`'s extractor (§2).
2. **The amplitude** is assembled from three papers and checked piece by piece (§3). One of them has a typo, and the charged-Higgs Barr–Zee term is built here.
3. **Exact S_3: zero** (§4).
4. **Soft vacuum: it removes about 70% of the scalar points** (§5).

The previous report found $h \to e\mu$ to be the binding LFV Higgs decay, excluding only 5 points. The reason $\mu \to e\gamma$ is so much stronger is standard [3]: a Barr–Zee diagram needs only one flavour-violating coupling, combined with the large hWW and htt couplings, and MEG II is more sensitive than any Higgs measurement.

2 Couplings of every scalar

$h \to \ell\ell'$ needed only the CP-even couplings. Here all three kinds enter. Expanding each doublet as $H_k = \big(c_k, (v_k + r_k + i a_k)/\sqrt{2}\big)$ in the five-term Yukawa Lagrangian and differentiating gives per-doublet matrices in the convention $L \supset -\bar{e}_L Y^{\phi} e_R\, \phi - \bar{\nu}_L Y_C e_R\, H^+ + \text{h.c.}$ of [2, 3]. The doublet structure fixes

$$Y^{(a_k)} = i\, Y^{(r_k)}, \qquad Y^{(c_k)} = \sqrt{2}\, Y^{(r_k)},$$

and the computation reproduces both relations. The physical couplings follow from the rotation columns. Unlike $h \to \ell\ell'$, the dipole needs the *signs* of the lepton eigenvalues, because its τ-enhanced term $m_\tau Y^*_{\tau e} Y^*_{\mu\tau}$ flips sign with the τ eigenvalue. The right-handed fields are rephased to make every mass positive.

check	against	agreement
all 7 scalars' 3×3 couplings, soft point	`feynlag` extractor	10^{-18}
Goldstone couplings flavour-diagonal	$i\,\mathrm{diag}(m)/v$, $\sqrt{2}\,\mathrm{diag}(m)/v$	10^{-15} (102 points)

The Goldstones carry no flavour violation at any vacuum. This is what allows them to be dropped (their part of the W loop is already inside the Barr–Zee W term).

3 The amplitude

With $T = e\,\varepsilon^*_\alpha \bar{u}_e\, m_\mu i\sigma^{\alpha\beta} q_\beta (A_L P_L + A_R P_R) u_\mu$,

$$\mathcal{B}(\mu \to e\gamma) = \frac{48\pi^3 \alpha\, (|A_L|^2 + |A_R|^2)}{G_F^2} \qquad [2], \text{ Eqs. (28)–(29).}$$

[3]'s Wilson coefficients are $c = 4\pi^2 A$, and the two papers' one-loop results agree under this map, sign included. They then print the same Barr–Zee formulae, so one-loop and two-loop pieces can be added with a common sign.

3.1 One loop

[3] Eq. (A3) gives the neutral-scalar diagram as a two-parameter Feynman integral. With the external masses dropped from the denominator ($m_\mu^2/m_\phi^2 \lesssim 10^{-6}$) it integrates in closed form:

$$c_L = \sum_{f=e,\mu,\tau} \frac{\frac{1}{2} I_1(t_f)\,(m_e Y_{ef} Y^*_{\mu f} + m_\mu Y^*_{fe} Y_{f\mu}) + I_2(t_f)\, m_f Y^*_{fe} Y^*_{\mu f}}{4 m_\mu m_\phi^2}, \qquad t_f = \frac{m_f^2}{m_\phi^2},$$

$$I_1 = \frac{t^3/6 - t^2 + t\ln t + t/2 + 1/3}{(1-t)^4}, \qquad I_2 = \frac{t^2 - 4t + 2\ln t + 3}{2(t-1)^3},$$

and c_R is obtained by $Y \to Y^\dagger$. This is exact in m_f/m_ϕ, so a light scalar needs no expansion. It agrees with the full integral, external masses kept, to 10^{-8} at $m_\phi = 700$ GeV and 6×10^{-6} at 25 GeV. For $t \to 0$ it gives the familiar $\ln(m_\phi^2/m_f^2) - \frac{3}{2}$.

A typo in [2]. Their Eq. (30) multiplies the non-flip term $y_{i\mu}/6$ by m_i/m_μ. That term takes its chirality flip from the external muon, so it carries m_μ, not m_τ. [3] Eq. (A3) and [2]'s own $\tau \to \mu\gamma$ formula agree with this.

The charged-scalar loop, with massless neutrinos, is $A_L = -(Y_C^\dagger Y_C)_{e\mu}/(192\pi^2 m^2_{H^\pm})$ and $A_R = 0$ [2]. This is $-\frac{1}{2}$ of the neutral non-flip term, and [5]'s general formulae give the same ratio independently.

3.2 Barr–Zee

The fermion and W loops in the $\phi\gamma\gamma$ vertex are [2] Eq. (20), which is identical to [3] Eqs. (A4)–(A5).

piece	reference value	this code
top, $\tau \to \mu\gamma$ via the SM Higgs	$-0.082\, Y_{tt}$ [3] Eq. (14)	-0.0811 (1%)
W, same	0.11 [3] Eq. (14)	0.1254 (14% above)
W bracket, $m_\phi = 125$–800 GeV	$F^{(3)}$ of [4]	1%

The W formula is the one both papers print, and it matches [4]'s independent $g-2$ calculation. The 14% gap is therefore with [3]'s *number*. The neglected $\phi Z\gamma$ piece moves it the wrong way (+6%), so the likely cause is [3]'s "set C" diagrams, which are not implemented. Once the gap is rescaled out, the full assembly reproduces [3]'s $\mu \to e\gamma$ bound on an SM-like Higgs to 2%. With MEG II that bound becomes

$$\sqrt{|Y_{\mu e}|^2 + |Y_{e\mu}|^2} < 7.2 \times 10^{-7}.$$

The charged-Higgs Barr–Zee term appears in neither [2] nor [3]. [4] gives both one-loop $\phi\gamma\gamma$ vertices with an off-shell photon. They have the same k^2-dependence, so the second loop is the same integral, and the $H^\pm$ term follows from the fermion term by the substitution $m_f y_{ff} N_c Q^2\,(1-2x(1-x)) \to (v\lambda/2)\,x(1-x)$. The sign of λ relative to `feynlag`'s trilinear ($L \supset g\,\phi H^+ H^-$, $\lambda = -g/v$) is not assumed. It is fixed by the low-energy theorem, which requires the scalar-to-top ratio of the heavy-loop $\phi\gamma\gamma$ vertices to equal that of `feynlag`'s validated diphoton amplitude.

Not included: the $\phi Z\gamma$ Barr–Zee diagrams (suppressed by $1-4s_W^2$), the $H^\pm W^\mp\gamma$ ones, and [3]'s set C. The quark couplings are an **ansatz**. Each quark's Yukawa lies along the vacuum direction, so $y_{tt}(h_i) = \kappa_i m_t/v$ and the CP-odd states have none, as in the previous report's $R_{\gamma\gamma}$.

4 The exact-S_3 vacuum: identically zero

At 40 exact points of the first scan, evaluated at $\varphi = \pi/3$ with no soft terms, at three values of μ_3^ℓ, the rate is $\mathcal{B} \sim 10^{-30}$, which is round-off. This is not because the couplings are small: h_0, one A and one $H^\pm$ carry $e\mu$ couplings up to ~ 0.1.

The reason is the **residual** $\mathbb{Z}_2$ of the aligned vacuum, the S_3 reflection that leaves it invariant. Under it the electron, which is the generation the O_{12} rotation decouples, is odd, while μ and τ are even. The scalars orthogonal to the vacuum *within* the doublet plane (h_0, one A, one $H^\pm$) are odd, and the rest are even. A $\mu \to e\gamma$ amplitude has one odd external lepton and is forbidden while the $\mathbb{Z}_2$ holds. In particular h_0 has neither a VV coupling nor a diagonal $h_0 H_a^+ H_a^-$ coupling, so its $e\mu$ coupling feeds no Barr–Zee diagram. This is the radiative-decay counterpart of the first-generation decoupling that forces $V_{us} = 0$ in the fermion-sector report.

So $\mu \to e\gamma$ **measures the soft breaking directly.** Moving one exact point off the alignment, with the free soft terms still zero, gives

$$\mathcal{B} \propto (60^\circ - \varphi)^2 \quad \text{(fitted slope 2.002)},$$

that is, an amplitude linear in the $\mathbb{Z}_2$ breaking. A misalignment of 0.1° already gives 1.9×10^{-13}, at the MEG II limit.

5 The soft-broken scan

Every point × the 73-value μ_3 grid of the previous report × every lepton branch gives 417 992 configurations. For each one the SM-like state's $h \to \ell\ell'$ rates are computed alongside, so the two constraints are compared configuration by configuration.

	configurations allowed	scalar points allowed
three Higgs LFV limits [6, 7]	79.8%	2023 / 2028
MEG II	2.51%	**618 / 2028**
both		606 / 2028

The median rate is $\mathcal{B} = 7.7 \times 10^{-10}$. Configuration by configuration, measured against its own limit, $\mu \to e\gamma$ is a median 9.2×10^4 times more restrictive than $h \to e\mu$. Unlike every Higgs LFV decay, it **constrains the scalar sector**.

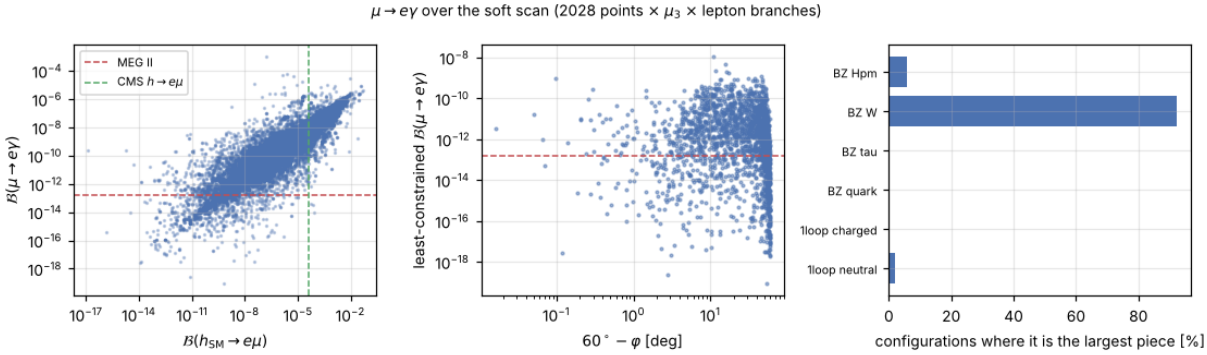


Figure 1: Left: $\mathcal{B}(\mu \to e\gamma)$ against the SM-like $\mathcal{B}(h \to e\mu)$ per configuration (40 000 sampled), with the MEG II and CMS limits dashed. Middle: the least-constrained configuration of each scalar point against the vacuum misalignment. Right: which piece of the amplitude is largest, by share of configurations.

What drives it. The Barr–Zee W loop through the SM-like Higgs is the largest piece in 92% of configurations, the $H^\pm$ Barr–Zee loop in 5.8%, and the one loop in about 2%. The SM-like state's $e\mu$ coupling is pure non-vacuum admixture (previous report), so the survivors are the more aligned points: their median $1-\kappa_V^2$ is 8.3×10^{-4}, compared with 2.4×10^{-3} overall. No μ_3 window or lepton-branch kind is singled out. MEG-allowed configurations span the whole $|\mu_3|$ axis and all three branch kinds. Being close to $\varphi = \pi/3$ does not protect a point (Fig. 1, middle). In the scan the free soft terms break the $\mathbb{Z}_2$ whatever φ is, so φ is not its order parameter.

Robustness. The two least certain inputs are the quark couplings (an ansatz, and the top loop interferes destructively with the W loop) and the 14% W-loop gap. Each configuration carries three variants built from the same amplitudes:

variant	scalar points allowed by MEG II
baseline	618
no quark Barr–Zee	504
quark couplings with opposite sign	453
W loop scaled to [3]'s 0.11	684

The count moves by about 25%, and the conclusion that MEG II removes most of the scalar sector does not.

6 Open questions

1. **Missing two-loop pieces:** [3]'s set C, and the $\phi Z\gamma$ and $H^{\pm}W^{\mp}\gamma$ Barr–Zee diagrams. The variants above suggest their effect on the point count is of the same order as the quark uncertainty, but they are not computed.

2. **The quark sector per point**, to replace the Barr–Zee quark ansatz. The same input is missing from $R_{\gamma\gamma}$.

3. **An order parameter for the residual** $\mathbb{Z}_2$ in the soft vacuum: a combination of soft terms rather than φ. It would turn the near-zero of §4 into a scan variable.

4. $\mu \to 3e$ **and** μ–e **conversion** share this dipole and add contact terms. $\tau \to \mu\gamma$ **and** $\tau \to e\gamma$ need only the flavour indices of `radiative.py` generalized.

References

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